```
!pip install pandas numpy matplotlib seaborn scikit-learn
```

```
Requirement already satisfied: pandas in c:\users\ai_lab\anaconda3\
lib\site-packages (2.3.3)
Requirement already satisfied: numpy in c:\users\ai_lab\anaconda3\lib\
site-packages (2.3.5)
Requirement already satisfied: matplotlib in c:\users\ai_lab\
anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: seaborn in c:\users\ai_lab\anaconda3\
lib\site-packages (0.13.2)
Requirement already satisfied: scikit-learn in c:\users\ai_lab\
anaconda3\lib\site-packages (1.7.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\
ai_lab\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\ai_lab\
anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\ai_lab\
anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\ai_lab\anaconda3\
lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\ai_lab\
anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: scipy>=1.8.0 in c:\users\ai_lab\
anaconda3\lib\site-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in c:\users\ai_lab\
anaconda3\lib\site-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\
ai_lab\anaconda3\lib\site-packages (from scikit-learn) (3.5.0)
Requirement already satisfied: six>=1.5 in c:\users\ai_lab\anaconda3\
lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.svm import SVC
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_csv(r"C:\Users\AI_LAB\Desktop\ML_lab 241501173\iris\
Iris.csv")

df.head()
```

```
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
Species
0   1            5.1           3.5            1.4           0.2  Iris-
setosa
1   2            4.9           3.0            1.4           0.2  Iris-
setosa
2   3            4.7           3.2            1.3           0.2  Iris-
setosa
3   4            4.6           3.1            1.5           0.2  Iris-
setosa
4   5            5.0           3.6            1.4           0.2  Iris-
setosa
```

```
print(df.info())
print(df.describe())
print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
None
               Id  SepalLengthCm  SepalWidthCm  PetalLengthCm
PetalWidthCm
count  150.000000     150.000000    150.000000     150.000000
150.000000
mean    75.500000       5.843333      3.054000       3.758667
1.198667
std     43.445368       0.828066      0.433594       1.764420
0.763161
min      1.000000       4.300000      2.000000       1.000000
0.100000
25%     38.250000       5.100000      2.800000       1.600000
```

```
0.300000
50%     75.500000       5.800000       3.000000       4.350000
1.300000
75%    112.750000       6.400000       3.300000       5.100000
1.800000
max    150.000000       7.900000       4.400000       6.900000
2.500000
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

```
le = LabelEncoder()
df["Species"] = le.fit_transform(df["Species"])

X = df.drop(columns=["Id", "Species"])
y = df["Species"]

X.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.3,
    random_state=42,
    stratify=y
)

print("Training data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)
```

```
Training data shape: (105, 4)
Testing data shape: (45, 4)
```

```
svm_model = SVC(kernel="linear")

svm_model.fit(X_train, y_train)
```

```
SVC(kernel='linear')
```

```
y_pred_svm = svm_model.predict(X_test)

svm_accuracy = accuracy_score(y_test, y_pred_svm)

print("SVM Accuracy:", svm_accuracy)
print("SVM Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))
```

```
SVM Accuracy: 0.9111111111111111
SVM Confusion Matrix:
[[15  0  0]
 [ 0 14  1]
 [ 0  3 12]]
```

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier(random_state=42)
```